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**Java SE 27 & JDK 27**

# Class CRC32

java.lang.Object
    java.util.zip.CRC32

**All Implemented Interfaces:**

Checksum

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```
public class CRC32
extends Object
implements Checksum
```

A class that can be used to compute the CRC-32 of a data stream.

Passing a `null` argument to a method in this class will cause a `NullPointerException` to be thrown.

**Since:**

1.1

## *Constructor Summary*

**Constructors**

| Constructor | Description |
|---|---|
| `CRC32()` | Creates a new CRC32 object. |

## *Method Summary*

**All Methods** | **Instance Methods**

| Modifier and Type | Method | Description |
|---|---|---|
| `long` | `getValue()` | Returns CRC-32 value. |
| `void` | `reset()` | Resets CRC-32 to initial value. |
| `void` | `update(byte[] b, int off, int len)` | Updates the CRC-32 checksum with the specified array of bytes. |
| `void` | `update(int b)` | Updates the CRC-32 checksum with the specified byte (the low eight bits of the argument b). |

| | | |
|---|---|---|
| void | **update**(ByteBuffer buffer) | Updates the CRC-32 checksum with the bytes from the specified buffer. |

**Methods declared in class Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

**Methods declared in interface Checksum**

update

## *Constructor Details*

### CRC32

```
public CRC32()
```

Creates a new CRC32 object.

## *Method Details*

### update

```
public void update(int b)
```

Updates the CRC-32 checksum with the specified byte (the low eight bits of the argument b).

**Specified by:**

update in interface Checksum

**Parameters:**

b - the byte to update the checksum with

### update

```
public void update(byte[] b,
                   int off,
                   int len)
```

Updates the CRC-32 checksum with the specified array of bytes.

**Specified by:**

update in interface Checksum

**Parameters:**

`b` - the byte array to update the checksum with

`off` - the start offset of the data

`len` - the number of bytes to use for the update

**Throws:**

`ArrayIndexOutOfBoundsException` - if `off` is negative, or `len` is negative, or `off+len` is negative or greater than the length of the array `b`.

## update

```
public void update(ByteBuffer buffer)
```

Updates the CRC-32 checksum with the bytes from the specified buffer. The checksum is updated with the remaining bytes in the buffer, starting at the buffer's position. Upon return, the buffer's position will be updated to its limit; its limit will not have been changed.

**Specified by:**

`update` in interface `Checksum`

**Parameters:**

`buffer` - the ByteBuffer to update the checksum with

**Since:**

1.8

## reset

```
public void reset()
```

Resets CRC-32 to initial value.

**Specified by:**

`reset` in interface `Checksum`

## getValue

```
public long getValue()
```

Returns CRC-32 value.

**Specified by:**

`getValue` in interface `Checksum`

**Returns:**

the current checksum value

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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